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Holocene Lakes from Ramlat as-Sab'atayn (Yemen) Illustrate the Impact of Monsoon Activity in Southern Arabia

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Paleoecology and paleohydrology of the Ramlat as-Sab'atayn (Southern Arabia) are reconstructed from a comparative study including sedimentology, mineralogy, stable isotope ratios of carbonates, and palynology of lacustrine sediments recovered from the al-Hawa depression. The section dates from 8700 to 7200 yr B.P. and records an early phase of flooding followed by distinct lacustrine development from 7800 to 7200 yr B.P., coeval with maximum activity of the Indian monsoon. Comparison of the pollen record with modern pollen deposition suggests that regional vegetation was then already of desert type and was related to strong seasonal trade winds. © 1998 University of Washington.

Key Words: **Holocene; paleolakes; Yemen; Southern Arabia; sedimentology; clay mineralogy; isotope stratigraphy; palynology.**

INTRODUCTION

Strong archaeological evidence (Inizan *et al.,* in press) demonstrates that Yemen has been settled by humans for a long time, particularly in the now-arid region of Ramlat as-Sab'atayn. The oldest ^{14}C age for Neolithic occupation is 8500 yr B.P., a time that coincides with the early Holocene humid maximum widely recorded through the tropics (e.g., Petit-Maire *et al.,* 1995). It corresponds also, in this area, to a peak of summer monsoon activity recorded between 8800 and 7800 yr B.P. in marine sediments of the Arabian Sea (Sirocko *et al.,* 1993). New pollen, sediment, and isotope data from lacustrine deposits in Yemen characterize this interval in southern Arabia, and provide evidence of associated vegetation and hydrology. Pollen data from this region are scarce and consist only of general information about the vegetation of the northernmost Rub' al-Khali desert of Saudi Arabia during Holocene humid phases (El-Moslimany, 1983). Poor preservation of the pollen did not permit reliable reconstructions of the vegetation to be made. However, pollen from soil surface samples obtained during the present study in the Ramlat as-Sab'atayn provides a basis for interpreting the Holocene sediments at two sites in the al-Hawa depression (15°52′N, 46°52′W, 710 m altitude).

MODERN GEOGRAPHICAL FEATURES

The Ramlat as-Sab'atayn Desert

The Ramlat as-Sab'atayn (ca. 14°40′–16° N lat., 45°30′–47°30′ E long., 700–900 m altitude) stretches as a vast triangular sand-dune field bounded to the south and east by the Jawl and Hadramawt calcareous plateaus and to the westsouthwest by the Yemen volcanic and metamorphic highlands (Fig. 1). To the north, a series of eroded inselbergs marks the transition to the Rub'al-Khali desert. Dune fields to the east and southeast are cut by Wâdîs from the adjacent highlands (e.g., Wâdî al-Jawl, W. Markha). The channels, which contain silt deposits, are fringed by bushes. Lacustrine sediments of Holocene and Pleistocene age crop out in large interdune depressions in the central part of the area, at al-Hawa (Inizan *et al.,* in press), and also at the foot of the Jawl plateau near Shabwa (Inizan and Ortlieb, 1987).

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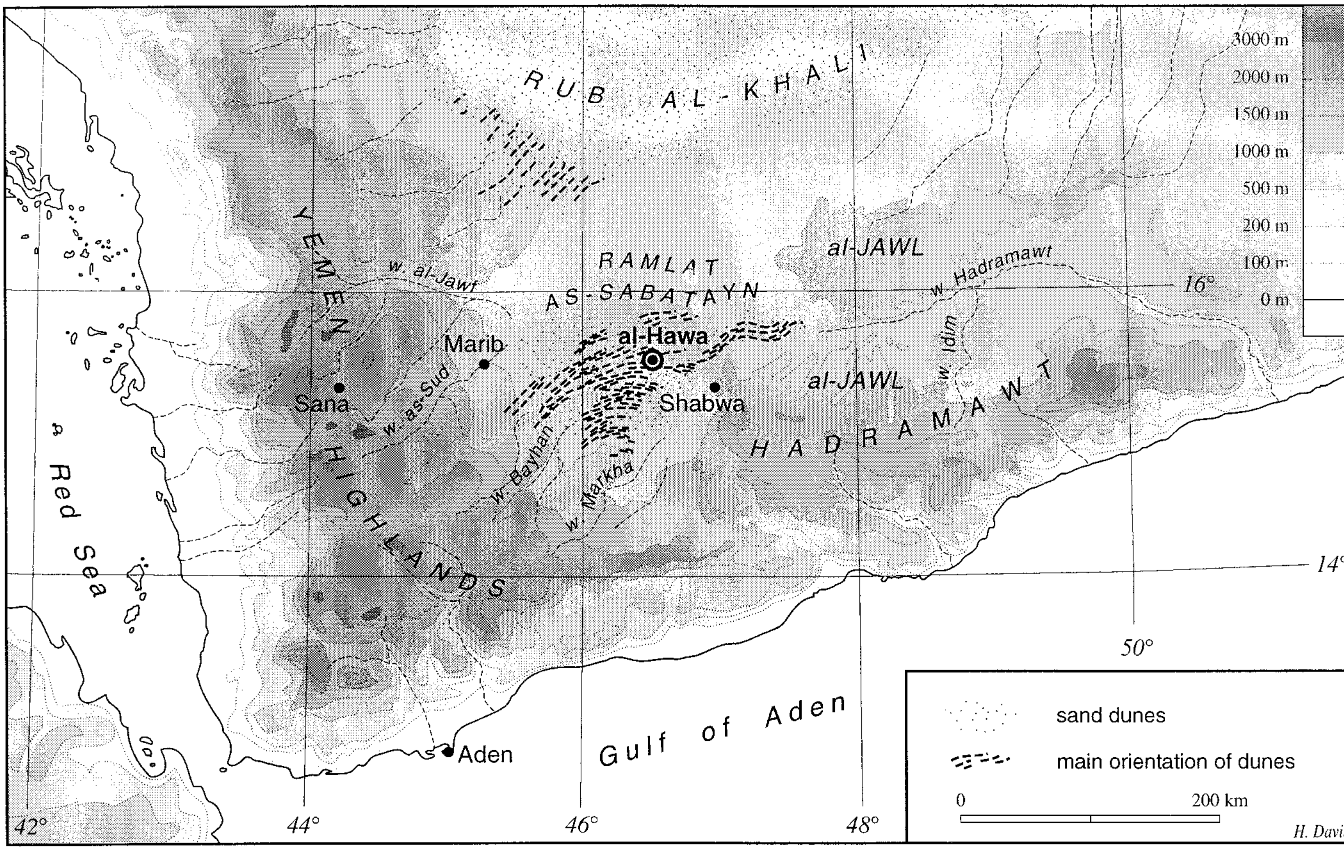

FIG. 1. Topographic map of the Ramlat as-Sab'atayn region in Yemen.

The Asian Monsoon and Regional Atmospheric Circulation Patterns

The regional climate is driven by the seasonal fluctuations of the Intertropical Convergence Zone (ITCZ). During summer, the differential sensible heating between land and ocean leads to the formation of low atmospheric pressure over the Indian continent and the slopes of the Himalayas, and high atmospheric pressure above the cooler Indian Ocean at about 30°S (Fig. 2). As a result, air flows strongly across the Indian Ocean along a SW–NE direction, and absorbs the available moisture. When moist air masses reach the southern coast of the Arabian Peninsula they rise along the highlands of Yemen and Hadramawt, where precipitation occurs. North of the highlands, the main part of the Arabian Peninsula lies under the influence of strong NW–SE trade winds, which are responsible for important dust transport. During winter, southward displacement of the ITCZ and rapid cooling over the Asian landmass induce a reversal of temperature and atmospheric pressure gradients, and allow dry NE monsoon winds to develop over the Arabian Peninsula, and dust to reach the Arabian Sea and the northern Indian Ocean (Sirocko *et al.,* 1993). As a result, the regional climate of the southern Arabian Peninsula is dry, with mean annual rainfall decreasing from west (390 mm/yr at Sana'a in the western highlands) to east and north. In the Ramlat as-Sab'atayn, precipitation does not exceed 200 mm/yr and occurs during the hottest months, when evaporation is intense; the precipitation/evaporation ratio averages 0.03 to 0.2 (Walter *et al.,* 1975).

Vegetation

Desert vegetation develops on sand dunes. It consists mainly of herbaceous taxa, including *Panicum turgidum, Leptadenia pyrotechnica, Calligonum comosum, Zygophyllum simplex,* and *Aristida plumosa* associated with *Dipterygium glaucum, Cyperus conglomeratus,* and annuals such as *Plantago ciliata* and *Anastatica hierochuntia.* Where present, trees are mainly *Acacia tortilis, A. oerfota,* and *A. hamulosa* associated with *Commiphora myrrha* in the foothills and *Tamarix nilotica* along the Wâdîs. Saline habitats are characterized by *Salsola imbricata* and *Suaeda aegyptiaca.* Scattered occurrences of *Boswellia sacra* are observed only on the plateau of Jawl (Bunker, 1953; El-Moslymani, 1983; Al-Hubaishi and Müller-Hohenstein, 1984; Wood, 1997).

MODERN POLLEN DEPOSITION

Six surface samples from soil and Wâdî mud of the Ramlat as-Sab'atayn semi-desert steppe and adjacent wooded area

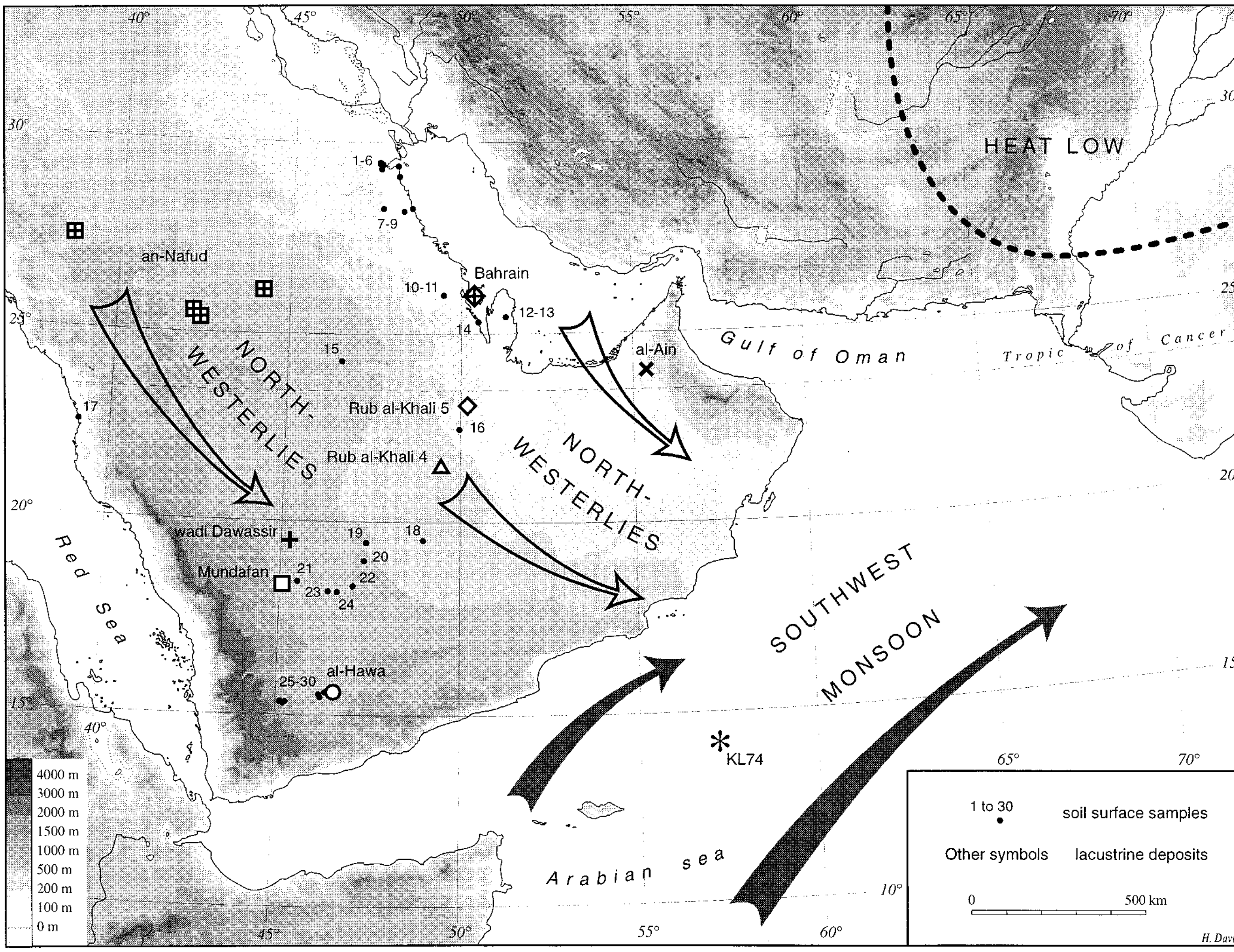

FIG. 2. Main geographic features of the Arabian Peninsula and location of samples mentioned in the text.

were analyzed. Related pollen assemblages (results expressed as percentages of the total, including all pollen and spore taxa identified) are shown in Figure 3, together with data from soil surface samples collected over the entire Arabian Peninsula up to southern Iraq (El-Moslimany, 1983; Bonnefille and Riollet, 1988) (Fig. 2). Samples were processed using the standard HF method (Faegri and Iversen, 1975) and sieved on a 5-μm mesh. Pollen counts ranged from 358 to 933, and 54 taxa have been identified (Table 1). They have been determined using the reference slide collection of the Museum National d'Histoire Naturelle in Paris and atlases of pollen morphology (e.g., Reille, 1992; Bonnefille and Riollet, 1980; El Ghazali, 1991).

Surface samples from the Ramlat as-Sab'atayn (IIb, samples 25–28) show the dominance of Cyperaceae (maximum 45%), *Tribulus* (18%), *Dipterygium* (17%), Chenopodiaceae (mainly *Aerva lanata*) (25%), and Gramineae (25%), which reflects the large extension of dune fields. *Blepharis, Artemisia,* Cruciferae, Compositae, and *Calligonum* may also occur locally in low percentages. Arboreal pollen (*Commiphora, Salvadora,* and *Acacia*) are always less than 1%.

Acacia increases in samples from wooded areas around Marib (zone IIc, samples 29–30), with percentages as high as 14% in sample 29. There, *Zygophyllum* and Chenopodiaceae reach 22% and supplant Cyperaceae and *Dipterygium,* which decrease to 10 and 5%, respectively.

These pollen spectra are similar to those recorded to the north in the Rub' al-Khali desert (El-Moslimany, 1983) (zone IIa, samples 18–24). However, the near absence of Gramineae (<2%), and significant percentages of *Limeum* and *Fagonia* (maxima 9% and 7%) in the Rub' al-Khali testify to more desertic conditions.

North of 22°N (zone I, samples 1–17), pollen assemblages are dominated by Chenopodiaceae, *Plantago,* and Compositae. These pollen types reflect different environmental conditions, with colder winter temperatures (El-Moslimany, 1983; 1990) and rainfall confined to the winter season. Gramineae are

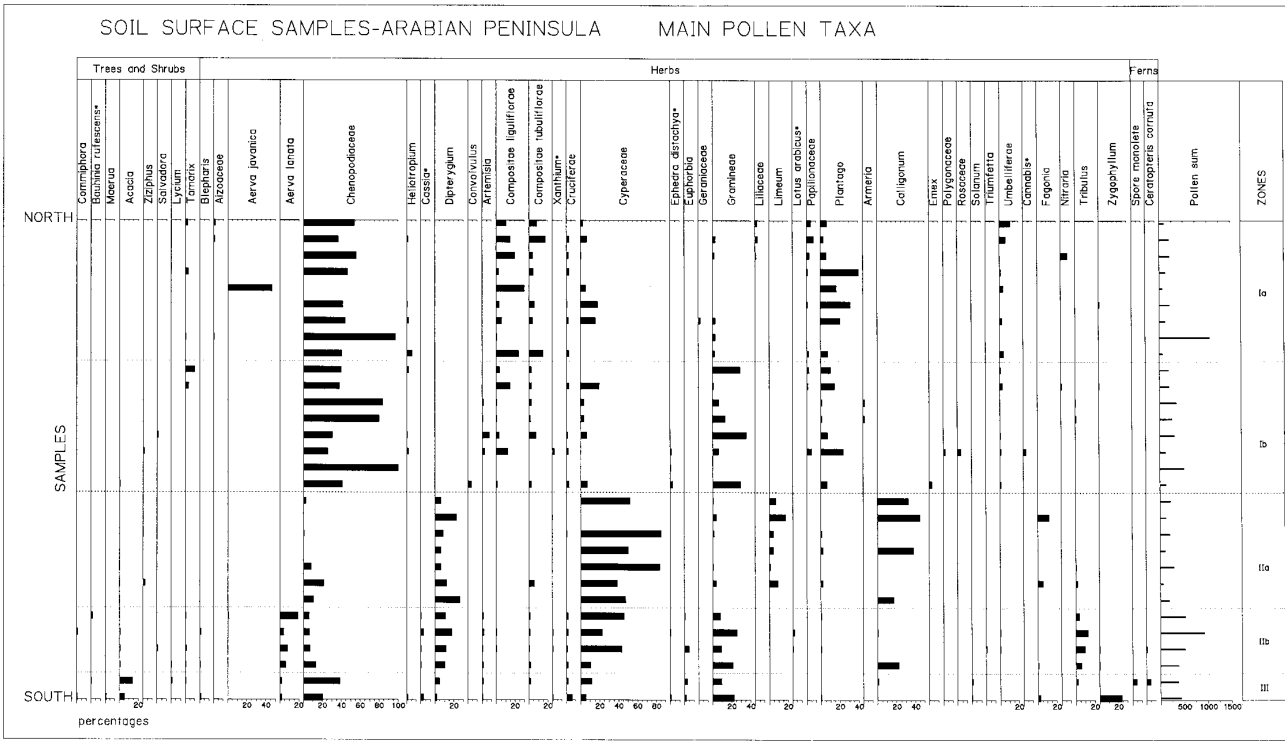

FIG. 3. Summary pollen diagram of modern soil surface samples in the Arabian Peninsula plotted from north (sample 1) to south (sample 30).

present in significant amounts in coastal areas (zone Ib, samples 10–17), but they decrease northward and are absent from southern Iraq and Kuwait (zone Ia, samples 1–9).

Modern pollen deposition in desertic and semi-desertic areas of Arabia clearly reflects major characteristics of the environment (i.e., temperature and rainfall regimes), and distinct influences from tropical and temperate areas.

THE AL-HAWA HOLOCENE RECORD

The lacustrine deposits were sampled in two sections (Hawa 2 and 4) a few meters apart in the same depression (Fig. 4). Two sedimentary units have been distinguished. Unit B consists of dark-gray to black, unconsolidated, abiotic calcareous silts, about 100 cm thick. A few thin layers are enriched in sand-sized or clay-sized particles. This unit was sampled at site Hawa 4. The overlying Unit A consists of alternating white-to-gray, consolidated, calcareous silts and diatomaceous calcareous silts. Some layers contain abundant remains of *Melania tuberculata.* This unit is 20 to 80 cm thick, and was sampled at both sites.

Radiocarbon Chronology

Radiocarbon measurements of organic (algal) matter at the base of Unit B and on shells of *Melania tuberculata* from the consolidated lacustrine deposits of Unit A yielded normalized (Stuiver and Polack, 1977) ages ranging from 8700 to 7200 yr B.P. (Table 2). They have been compared to ^{14}C analyses of the total silt-sized calcite fraction of the sediment to estimate input of detrital carbonate from the Tertiary carbonate bedrock that crops out in the drainage basin of the al-Hawa depression. Very similar results were obtained from both sets of samples (<2% of dead carbon), indicating that contamination is very low for ^{14}C measurements and negligible for ^{13}C measurements. This suggests that detrital input of calcareous particles from the drainage basin is insignificant. However, dissolved calcium and bicarbonate may have been transported to the depression by running water. We therefore assume that the calcite particles represent 65 to 90% of the bulk sediment formed *in situ* in the lacustrine environment, with a probable pH near or above 8.5 (Fontes and Gasse, 1989).

The ^{13}C content of calcite particles ranges from +2.49 to −0.57‰ (δ vs PDB standard). The high, positive values indicate that carbonate precipitation occurred near the surface, in equilibrium with ambient atmospheric CO_2 for water temperatures of about 25°±5°C (Emrich *et al.*, 1970). The lowest, negative, values probably result from the respiration of the *Typha* population that developed in swamp areas around the lake during phases of low water, as indicated by high percentages of related pollen grains in these samples (at 40 and 95 cm depth). By comparison, the ^{13}C amounts measured in *Melania* shells, corrected for aragonite–calcite fractionation (Rubinson and Clayton, 1969; Tarutani *et al.*, 1969), deviate systemati-

TABLE 1

Pollen Taxa Determined in the Modern Soil Surface Samples from Yemen (1) and in the al-Hawa Holocene Sequence (2)

Local and regional Plants		Extra-regional plants
Arboreal pollen	Non-arboreal pollen	arboreal pollen
Acacia (1–2)	Herbaceous plants	Temperate
Commiphora (1–2)	*Aerva javanica* (1–2)	*Pinus* (1)
Lycium (1)	*Aerva lanata* (1–2)	*Betula* (2)
Maerua[a] (1–2)	*Artemisia* (1–2)	*Cedrus* (2)
Rhamnaceae (2)	*Blepharis* (1)	*Fagus* (2)
Salvadora (1)	*Calligonum* (1)	*Quercus* (deciduous) (2)
Tamarix (1)	*Cassia*[a] (1–2)	Tropicals
	Caryopyllaceae (1–2)	*Bauhinia rufescens*[a] (1)
	Centaurea perrottetii[a] (2)	Caesalpiniaceae (2)
	Celosia (1)	*Dodonaea* (1–2)
	Chenopodiaceae (2)	*Juniperus* (2)
	Commelina benghalensis[a] (1)	
	Compositae tubul. (1–2)	
	Compositae ligul. (1–2)	
	Convolvulaceae (2)	
	Cruciferae (1–2)	
	Cyperaceae (1–2)	
	Dipterygium (1–2)	
	Ephedra (1–2)	
	Euphorbia hypericifolia[a] (1)	
	Euphorbiaceae (1)	
	Fagonia (1–2)	
	Gnidia (1)	
	Gramineae (1–2)	
	Heliotropium steudneri[a] (1)	
	Hypoestes[a] (1)	
	Justicia (1)	
	Justicia anselliana[a] (1)	
	Labiatae (1–2)	
	Lotus arabicus[a] (1)	
	Papilionaceae (1)	
	Plantago (1–2)	
	Polygala (1–2)	
	Polycarpaea (1)	
	Resedaceae (1–2)	
	Rubiaceae (1)	
	Rumex (2)	
	Solanum (1)	
	Suaeda[a] (1)	
	Tribulus (1–2)	
	Umbelliferae (1)	
	Urticaceae (2)	
	Triumfetta (1)	
	Typha (1–2)	
	Veronia perrottetii[a] (1)	
	Xanthium[a] (1)	
	Zygophyllaceae (1)	
	Zygophyllum[a] (1–2)	
	Ferns and Mosses	
	Azola (2)	
	Ceratopteris[a] (1–2)	
	Pteris[a] (2)	
	monolete (1–2)	
	trilete (2)	
	Anthocerothaceae (2)	

[a] Pollen types grouping several genera or species.

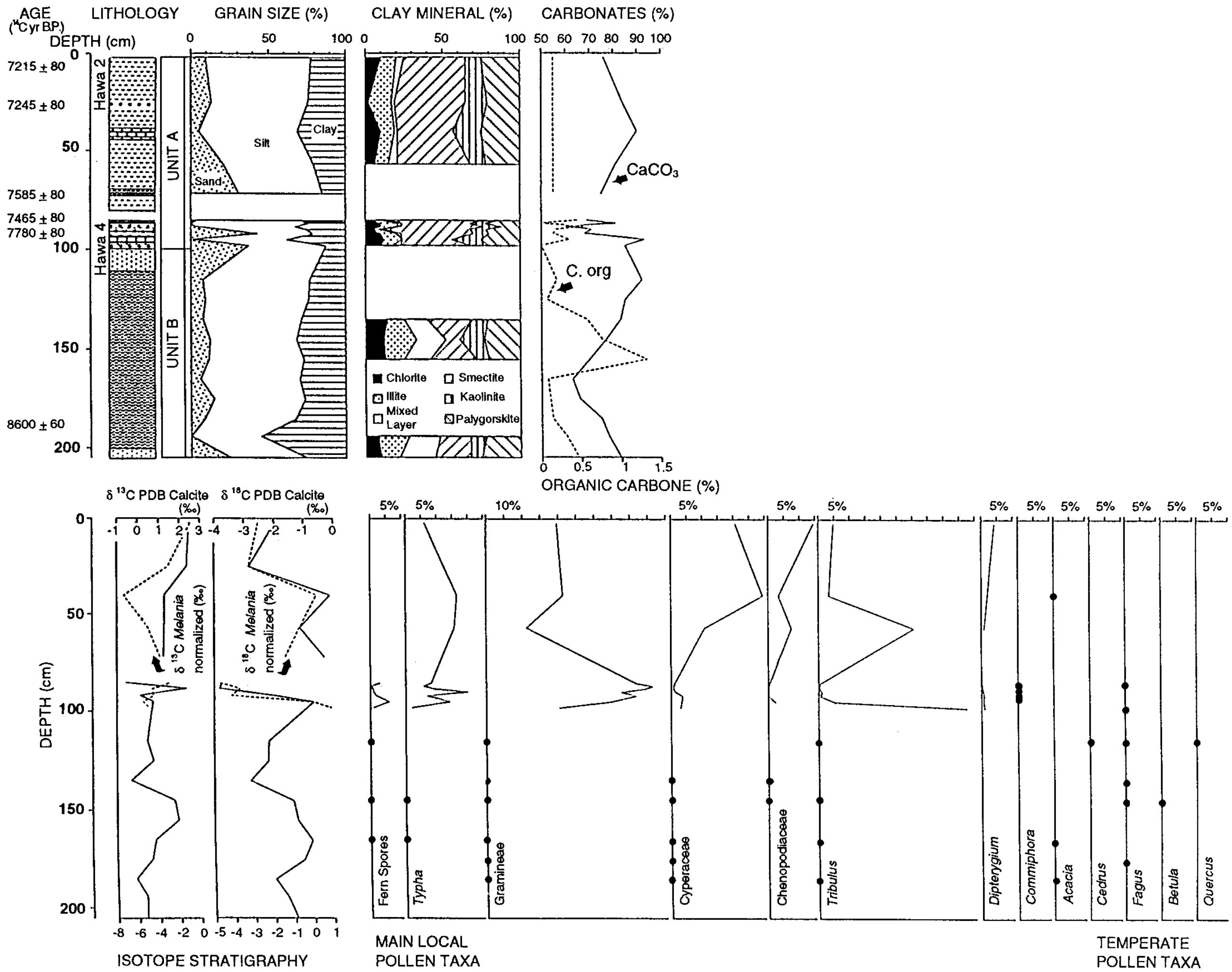

FIG. 4. The al-Hawa Holocene sequence (15°52′N, 46°52′W, 710 m alt.) depicting lithology, sedimentology, clay mineralogy, isotope stratigraphy, and palynology.

cally toward lower values and probably record the fractionation of the species (Fritz and Poplawski, 1974).

Paleohydrological Changes

Unit B is characterized by the large dominance of the angular to sub-angular, silt-sized particles of authigenic carbonate (calcite) described above. Very similar authigenic particles of calcite are common in modern lacustrine environments from temperate and tropical areas (Stabel, 1986; Talbot and Allen, 1996). Clay-sized particles are abundant in this unit (25 to 55% of the sediment), but they mostly consist of calcite. The scarcity of clay minerals in the clay fraction of some samples suggests that although some precipitation did occur, it was not enough to allow significant chemical weathering in the drainage basin. When clay minerals are present, the association consists of chlorite (10 to 15%), illite (10 to 20%), random mixed-layer clays (15 to 25%), kaolinite (5 to 15%), smectite (20 to 30%), and palygorskite (20 to 25%). Chlorite and illite are typical of physical weathering conditions and are ubiquitous in desert areas. They are associated with random mixed-layer clays, which originate from poorly weathered substrates (Chamley, 1989). These minerals could come either from dust transport by northerly winds (Sirocko *et al.*, 1991) or from erosion of the adjacent drainage basin.

Kaolinite develops in areas of high precipitation and good drainage (Chamley, 1989). Its distribution in Holocene sediments of the Arabian Sea suggests some transport by rivers to the Gulf of Oman (Sirocko and Lange, 1991). Therefore, some kaolinite may have developed in the drainage basin of the al-Hawa Trough during wet intervals; this is supported by its local association with rare diatoms in the sediment. Presence of palygorskite (which develops in sebkha-like environments) together with minerals derived from intense erosion of substrates suggest that it did not form close to the site at that time. Palygorskite is abundant in rocks and sediments that crop out in central Arabia, where it is associated with smectite; both

TABLE 2
Radiocarbon Ages and Isotope Measurements for the al-Hawa Holocene Sequence

	Depth (cm)	Laboratory number	Material	Age (^{14}C yr B.P.)	$\delta^{13}C$ calcite (‰ PDB)	$\delta^{18}O$ calcite (‰ PDB)	$\delta^{13}C$ *Melania* (‰ PDB)	$\delta^{18}O$ *Melania* (‰ PDB)	$\delta^{13}C$ *Melania* normalized	$\delta^{18}O$ *Melania* normalized	Water (25°C) *Melania* (‰ SMOW)
UNIT A	2	Pa 1533	*Melania*	7215 ± 80	2.47	−1.95	0.86	−2.19	−0.94	−2.79	−0.77
Hawa 2					2.49	−1.66	1.24	−2.01	−0.56	−2.61	
	25	Pa 1531	*Melania*	7245 ± 80	2.36	−2.83	−1.35	−2.69	−3.15	−3.29	−1.43
							0.2	−2.82	−1.6	−3.42	
	40				1.27	−0.1	−5.55	0.75	−7.35	0.15	1.84
							−3.78	0.29	−5.57	−0.31	
	57				1.27	−1.13	−3.21	−0.13	−5.01	−0.73	1.24
							−4.78	−0.03	−4.78	−0.63	
	72	Pa 1532	*Melania*	7580 ± 80	1.23	−0.3	−2.06	−0.84	−3.86	−1.44	1.62
					0.32	0.49	−1.62	−0.55	−3.42	−1.15	
Hawa 4	85.5	Pa 1559	*Melania*	7465 ± 80	−0.57	−3.77	−1.42	−4.14	−3.02	−4.74	−3.05
		Pa 1628	Calcite	7800 ± 60			−1.96	−4.63	−3.76	−5.23	
	88.5				2.34	−3.82	−2.84	−3.13	−4.64	−3.73	−1.79
							−2.11	−3.21	−3.89	−3.81	
	92				0.11	−1.95	−2.72	−3.55	−4.52	−4.15	−2.28
					0.7	−1.68	−3.04	−3.68	−4.85	−4.28	
							−2.57	−3.58	−4.37	−4.18	
	95	Pa 1567	*Melania*	7780 ± 80	0.73	−0.66	−3.69	0.67	−5.49	−0.07	1.85
	98						−3.12	1.5	−4.92	0.9	
UNIT B	115				0.47	−2.19					
	125				0.76	−2.21					
	135				−0.31	−2.81					
	145				1.77	−1.36					
	155				1.95	−1.22					
	165				0.84	−0.73					
	175				0.66	−1.02					
	185	Pa 1631	Calcite	8600 ± 60	−0.11	−2					
	194	Pa 1580	M.O.	8700 ± 100	0.38	−1.58					
	205				0.38	−1.26					

Note. The $\delta^{18}O$ content of lake water at 25°C temperature has been calculated using $\delta^{18}O$ values of *Melania* shells corrected for the aragonitic/calcite fractionation (t°C $= 16.9 - 4.2(\delta c - \delta l) + 0.13\,(\delta c - \delta l)^2$, with δc: $\delta^{18}O$ carbonate and δl: $\delta^{18}O$ water at the time of precipitation.

minerals are eroded and transported by northerly winds to the Arabian Sea (Sirocko and Lange, 1991). This indicates significant eolian contribution to the sediment of the al-Hawa depression.

Poor preservation of pollen grains in these levels suggests alternating phases of flooding and dessication. However, occurrences of taxa from northern, temperate, and montane ecosystems, like *Cedrus, Quercus, Betula,* and *Fagus,* corroborate the interpretation based on sedimentological data and point to noticeable trade wind activity from 8700 to 7800 yr B.P.

We interpret unit B as reflecting an early phase of precipitation and flooding ca. 8700 yr B.P., linked to high rainfall over the adjacent highlands. This interval marks an increase in the Indian summer monsoon strength, as recorded in Arabian Sea sediments at the same time (Sirocko *et al.,* 1993). However, northerly winds were strong during this period, and dry conditions probably prevailed temporarily, leading to successive phases of flooding and dessication.

Unit A is characterized by the presence of abundant diatoms, and occasionally by layers enriched in *Melania.* We interpret this as evidence of sedimentation in a perennial lake having relatively dilute water. The lake probably underwent fluctuations in level, as deduced from the pollen content: abundant fern spores and *Typha* alternate with *Tribulus* and Chenopodiaceae, indicative of a more arid and halophytic environment. Instability of hydrologic conditions is also recorded by the opposing fluctuations of ^{18}O and ^{13}C in the carbonates. The $\delta^{18}O$ measurements have been corrected to take account of the aragonite–calcite fractionation; similar values are found in calcite and *Melania* shells. The water content in $\delta^{18}O$, as calculated according to Craig (1965) for a water temperature of 25°C, varied between −3.05 and +2.8‰ SMOW. The rise of $\delta^{18}O$ in the levels characterized by the highest percentages of *Typha* and the lowest $\delta^{13}C$ values express strong evaporation of the lake during low-level phases.

Concomitant increases of diatoms and clay minerals and a decrease of carbonates in the lower part of Unit A most probably express a dilution effect, the consequence of intensi-

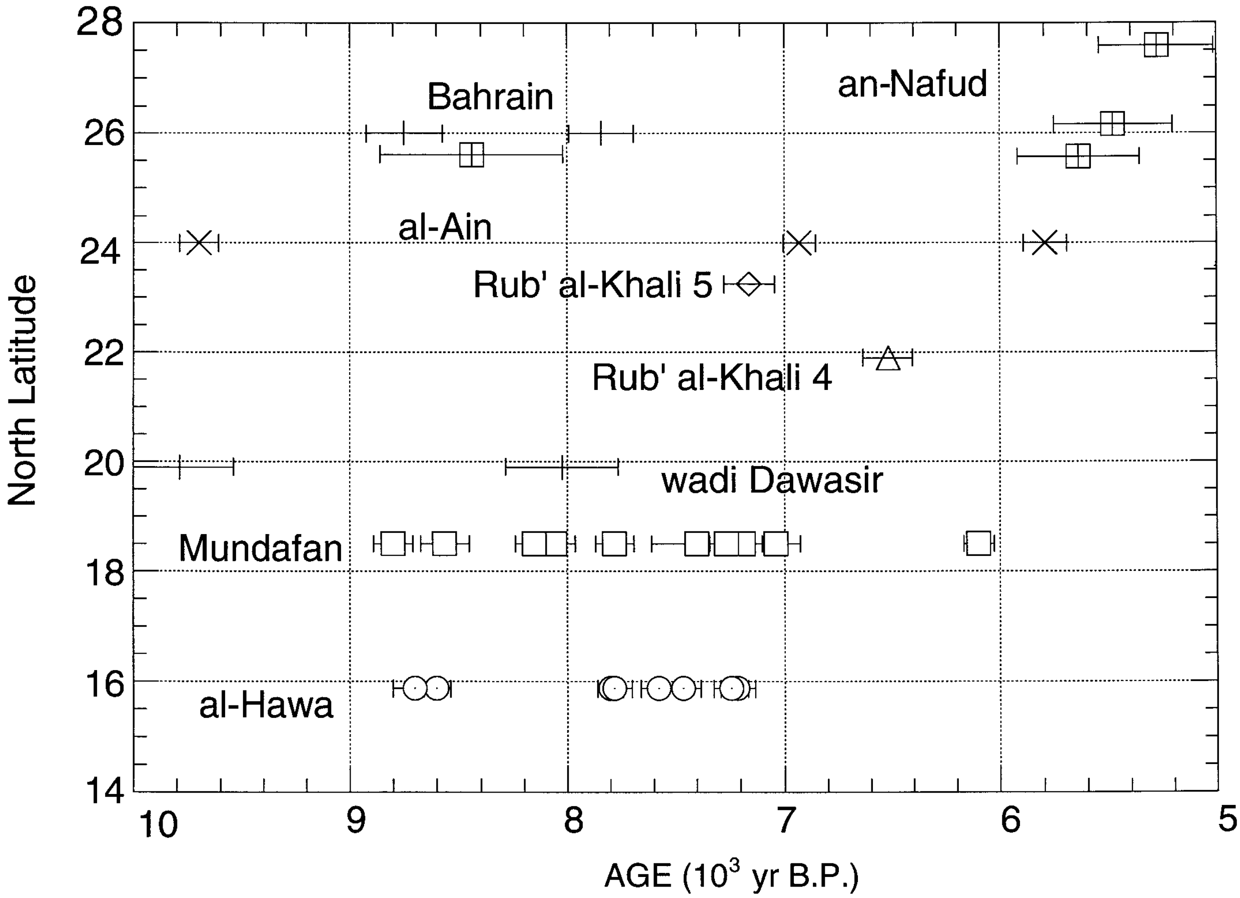

FIG. 5. ^{14}C measurements for the Holocene lacustrine sediments of Arabia plotted as a function of latitude.

fied chemical weathering and enhanced aquatic productivity. This, in turn, implies an important increase of humidity in southern Yemen. However, significant erosion of substrates persisted in the drainage basin during deposition of this unit. By comparison with Unit B, the clay mineral association of Unit A contains lower abundances of minerals typical of physical weathering and poor chemical weathering, chlorite (5 to 10%), illite (5 to 15%), and random mixed-layer clays (0 to 5%). Lower amounts of palygorskite (10 to 25%) indicate a decreased contribution of eolian sediment. The clay association is also characterized by increased amounts of minerals associated with chemical weathering processes. Higher contents of kaolinite (10 to 20%) indicate local but persistent formation of soils in upstream areas of the drainage basin; a significant increase in smectite (30 to 60%) at a time of decreasing eolian dust influx suggests proximal development of the mineral. Smectite probably formed through pedogenic processes and/or authigenesis in poorly drained, downstream areas of the drainage basis (Chamley, 1989).

Following emergence, lacustrine deposits were subjected to erosional processes. It therefore is likely that the lacustrine environment lasted for some time after 7200 yr B.P., the age of the upper outcropping sediment.

Regional Vegetation

The pollen content of Unit A is characterized by the dominance of herbaceous taxa (*Tribulus, Dipterygium*, Cyperaceae, and Gramineae). This pollen assemblage is very similar to the modern pollen association recorded in the Ramlat as-Sab'atayn region on unstabilized sand dunes (Fig. 3, zone IIb). It shows that the regional vegetation remained of desert type all through the period considered here (8700 to 7200 yr B.P.). Scattered occurrences of *Acacia* and *Commiphora* demonstrate that arboreal cover was present, and probably more important than it is now. However, the pollen spectra from early Holocene lacustrine deposits of Yemen do not indicate changes of vegetation as important as those described from correlative sediments in northern and western Africa, where Sudano-Sahelian wooded grasslands expanded widely along the southern margin of the Sahara (e.g., Lézine *et al.*, 1990; Ritchie and Haynes, 1987).

GENERAL DISCUSSION AND CONCLUSION

The sedimentary sequence of Al-Hawa formed during an interval of increased humidity in Yemen, coeval with maxi-

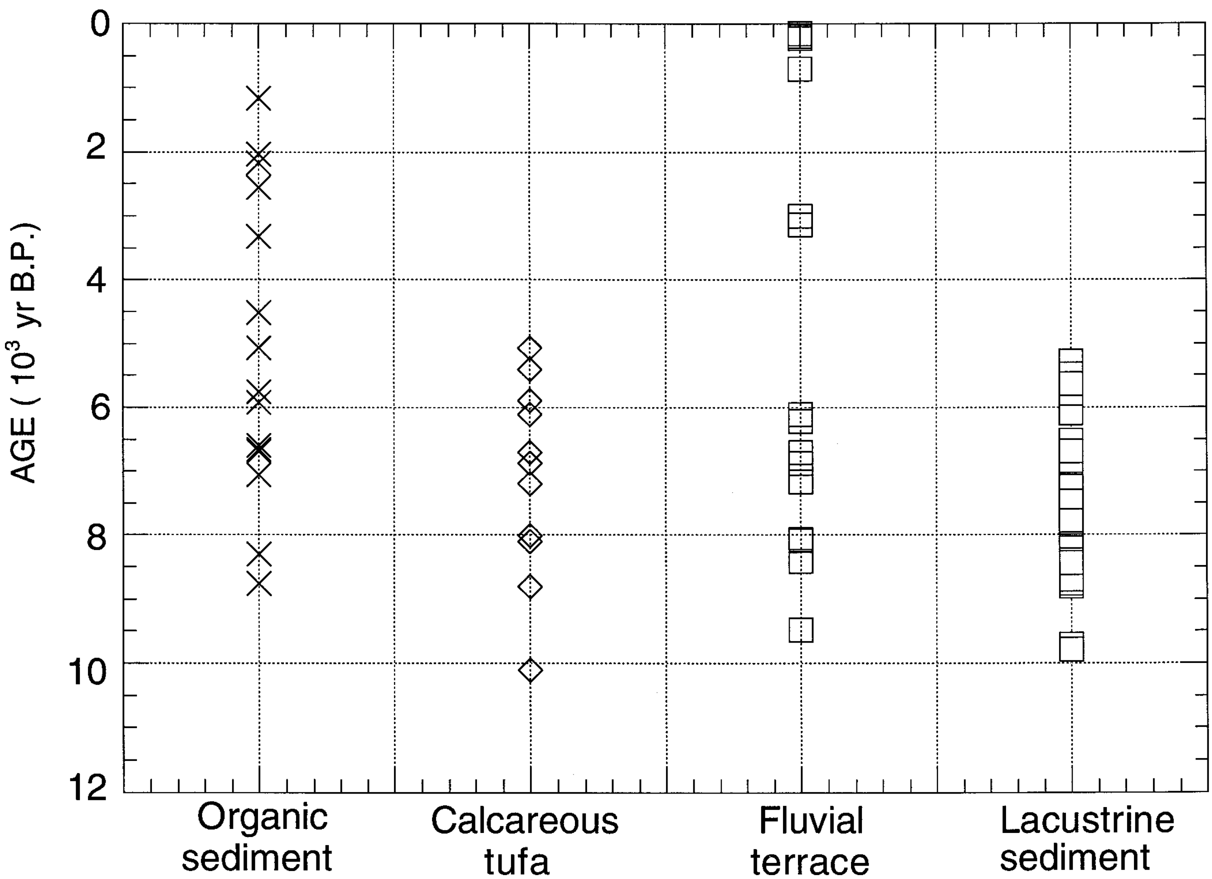

FIG. 6. Dated limnological records in Arabia between 10,000 yr B.P. and the present. Calcareous tufa from Oman (Clark and Fontes, 1990) have not been included in this general reconstruction, primarily due to the specific and local character of these records from hyperalkaline ground waters.

mum activity of the Indian summer monsoon during the early Holocene. Increased and/or more frequent rainfall near the transition from Unit B to Unit A (ca. 8000 yr B.P.) led to the development of a permanent lacustrine environment which lasted for at least 500 years, and probably more. This was associated with regional extension of chemical weathering, especially in the lower part of the drainage basin and/or the flood plain.

Similar, extensive lacustrine environments have been described in central Arabia in the Rub' al-Khali desert from 8800 to 6100 yr B.P. (Mc Clure, 1976) and at Wâdî Dawassir about 9800 yr B.P. (Zarins *et al.,* 1979) (Fig. 5), and in eastern Arabia at al-Ain from 9700 to 5800 yr B.P. (Gebel *et al.,* 1989). The lacustrine episode recorded in the northern an-Nafud desert between 25°37′ and 27°36′ N, lasted until 5300 yr B.P. (Whitney *et al.,* 1983; Schulz and Whitney, 1986), i.e., about 1000 years later than those recorded in the Rub' al-Khali and in the Ramlat as-Sab'atayn. The Mediterranean likely played a role in the climatic evolution of the northern part of the Arabian Peninsula.

Abundant travertines, fluvial terraces, and paleosols exposed widely throughout the Arabian Peninsula (e.g., Sanlaville, 1992; Al-Sayari and Zötl, 1978; Jado and Zötl, 1984; Roberts and Wright, 1993) provide additional evidence of regional humidity induced by intensified monsoonal precipitation fluxes over Arabia during the early Holocene. Related ^{14}C ages are shown in Figure 6. Those calculated from initial ^{14}C content measurements of 85 and 100% (Jado and Zötl, 1984) generate significant age uncertainties and are not shown. Figure 6 shows that this humid climatic episode is widely recorded in "lacustrine sediments" and in "calcareous tuffa," although samples for ^{14}C measurements have been taken from discontinuous or isolated sequences. On the contrary, "organic sediments" (paleosols) and "fluvial terraces" do not yield a precise paleoclimatic signal, for they may have recorded minor wet phases and/or seasonal runoff.

Whitney *et al.* (1983) have suggested that this early Holocene humid maximum coincided with semi-arid climate in the an-Nafud area, where a mean annual rainfall was about 235–300 mm, i.e., significantly greater than modern values (50–100 mm). Our data underline the role of northerly trade winds in the persistence of a sub-desertic regional environment in Yemen. This vegetational environment was not significantly different from modern conditions during all of the lacustrine episode. Additional data are needed to assess this observation at higher resolution, and to find additional links between the development of monsoonal activity, as recorded in the Indian Ocean at about 14,300 yr B.P. (Sirocko *et al.*, 1993), and the response of continental climates of the Arabian Peninsula.

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